70 Grade 12 Functions and Inverses

Questions:

1) f(x) = 2x-6

Determine the inverse equation of f(x)?

2) f(x) = $(x-4)^2 - 5$

Determine the inverse equation of f(x)?

3) f(x) = $2^x - 4$

Determine the inverse equation of f(x)?

4) f(x) = $\frac{2}{x-4} - 2$

Determine the inverse equation of f(x)?

5) f(x) = $(x-2)^2 - 5$

Draw the inverse of f(x)?

6) f(x) = $2^x - 2$
Draw the inverse of f(x)?

7) f(x) = log_2x
Determine the inverse equation of f(x)?

8) f(x) = log_2x
Draw the inverse of f(x)?

9) f(x) = $x^2 - 3x - 4$
Draw the inverse of f(x)?

10) $f(x) = 3^{-x}$
Determine the inverse equation of f(x)?

11) $f(x) = \log_2 x$
Draw the inverse of f(x)

12) $f(x) = \log_2 x$
Determine the equation of g(x) which is a reflection of f(x) in the x axis?

13) $f(x) = \log_2 x$
Determine the equation of g(x) which is a reflection of f(x) in the y axis?

14) $x = 3^y$
Solve for y?

15) $x = 2^{y-2} - 1$
Solve for y?

16) $b = log_a y$
Solve for y?

17) $log_x 8 = 3$
Solve for x?

18) $2^{x+3} - 2^x = 42$
Solve for x?

19) $7^{x-3} = 14$
Solve for x?

20) $3^{2x} - 11.3^x + 18 = 0$
Solve for x?

21) f(x) = $\frac{1}{2}^{x}$
Determine the inverse equation of f(x)?

22) g(x) = $-\log_5 x$
Determine the inverse equation of g(x)?

23) h(x) = $\log_4 (-x)$
Determine the inverse equation of h(x)?

24) h(x) = 4^{-x}
a) Draw h
b) Draw h^{-1}

25) h(x) = $\log_2 (-x)$
a) Draw h
b) Draw h^{-1}

26) f(x) = $3x^2$
Sketch f

27) f(x) = $-2x^2 + 4$ for $x < 0$
Sketch f

28) r(x) = 3x – 3
Determine the equation of g, the reflection of r(x) over the x axis.

29) f(x) = $\frac{2}{3}x^2$
Determine the equation of g, the reflection of f(x) across the line y = 0

30) f(x) = $\log_3 (x-1)$
Determine the equation of g, the reflection of f(x) across the line y = 0

31) f(x) = $-x^2 + 4$; $x \leq 0$
Draw f(x)

32) $f(x) = -2x^2$
Draw f(x)

33) $\log_3 x = 2$
Solve for x?

34) $5.4^{3x-3} = 50$
Solve for x?

35) $3.2^{x+1} = 20$
Solve for x?

36) $\log_3 x = 4$
Solve for x?

37) $\log x = 0$
Solve for x?

38) $\log_x 20 = 3$
Solve for x?

39) $2^{x-3} = 7$
Solve for x?

40) $8 = 3^{x-2}$
Solve for x?

41) $2\log_7 3x = 1$
Solve for x?

42) $f(x) = \sqrt{4x}$
Determine the domain of f(x)?

43) $h(x) = 4^{-x}$
Determine the equation of h^{-1}

44) $h(x) = \log_2(-x)$
Determine the equation of h^{-1}

45) $f(x) = 3x^2 - 9$
a) Sketch f^{-1}
b) Is f^{-1} a function?

46) $f(x) = 3x^2$
Determine the equation of f^{-1}

47) $r(x) = 3x - 3$
Determine the equation of g, the reflection of r(x) over the y axis.

48) $f(x) = \frac{2}{3}x^2$
Determine the equation of h, the reflection of f(x) across the line x = 0

49) $f(x) = \frac{2}{3}x^2$
Determine the equation of t, the reflection of f(x) across the line y = x

50) $f(x) = \log_3(x-1)$
Determine the equation of h, the reflection of f(x) across the line x = 0

51) $f(x) = \log_3(x-1)$
Determine the equation of r, the reflection of f(x) across the line y = x

52) $f(x) = -x^2 + 4$; $x \leq 0$
a) Draw f
b) Draw f^{-1}
c) Is f^{-1} a function?

53) $f(x) = \sqrt{4x}$
Determine the range of f(x)?

54) f(x) = $\sqrt{4x}$
a) Determine the inverse equation of f(x)
b) Draw the inverse equation
c) Draw f

55) g(x) = -2^x
a) Draw g
b) Draw g^{-1}

56) h(x) = $\log_3 x$
Draw h

57) g(x) = 3^x -3
Determine the y- intercept of g^{-1}

58) g(x) = 3^x -3
Determine the equation of g^{-1}?

59) f(x) = $\frac{1}{2}^{x}$ -2
a) Draw f
b) Draw f^{-1}

60) f(x) = $log_{\frac{1}{3}}x$
Draw f

61) f(x) = $\frac{1}{2}^{x}$ -2
Determine the equation of f^{-1}

62) f(x) = $\frac{1}{2}^{x}$ -2
Determine the equation of g(x) which is the reflection of f(x) across the line y = 0

63) $f(x) = \frac{1}{2}^{x} - 2$
Determine the equation of g(x) which is the reflection of f(x) across the line x = 0

64) $7^{x+3} = 4$
Determine x?

65) $7 = \log_x 3$
Determine x?

66) $2^{x+3} - 2^x = 5$
Determine x?

67) $f(x) = (x + 3)^2 - 4$
Determine the equation of f^{-1}

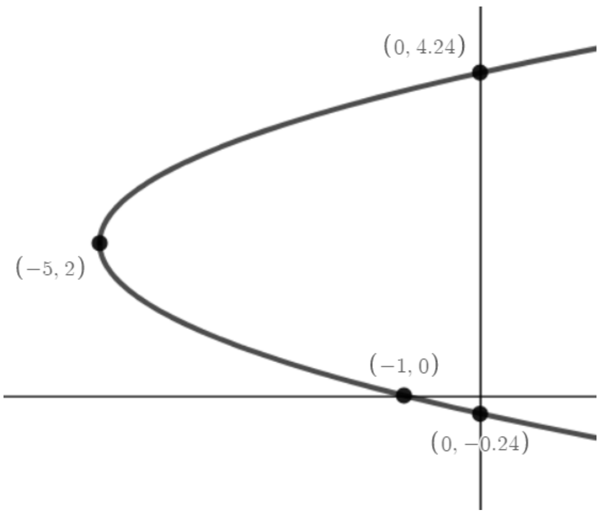
68) $f(x) = \frac{2}{x-4} - 4$
Determine the equation of f^{-1}

69) $f(x) = \frac{1}{4}^{x} - 4$
Draw f^{-1}

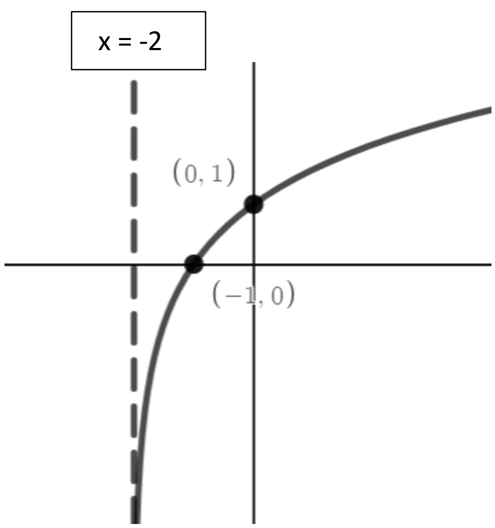
70) $f(x) = -\log_{4x}$
Determine the equation of f^{-1}

Answers:

1. $y = \frac{1}{2}x + 3$
2. $y = \pm\sqrt{x+5} + 4$
3. $y = log_2(x+4)$
4. $y = \frac{2}{x+2} + 4$
5.



6.



7. $y = 2^x$

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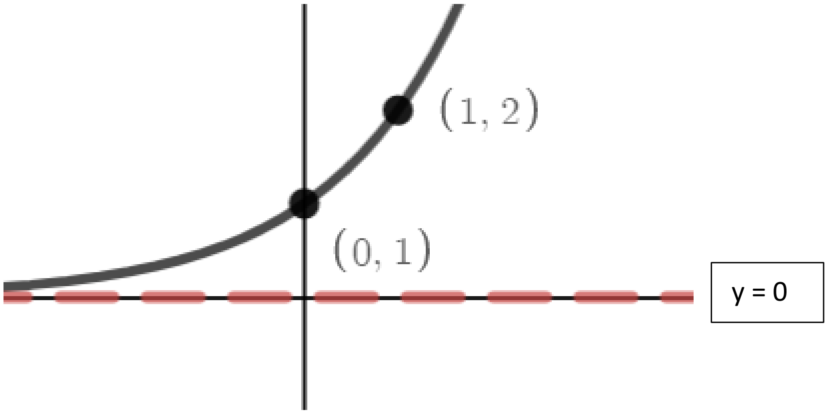

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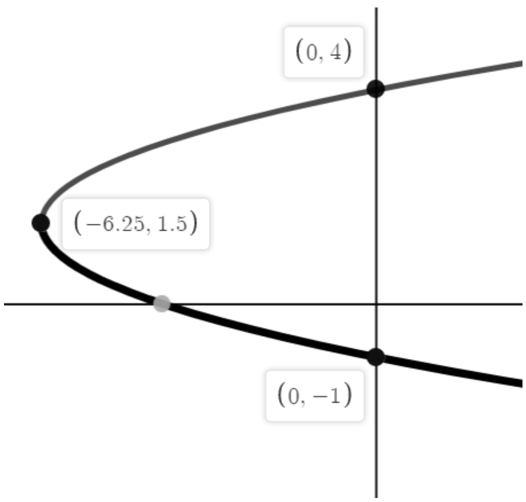
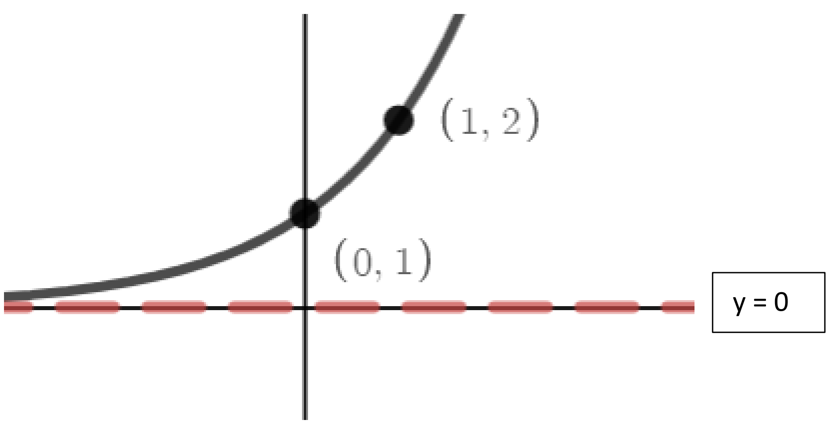
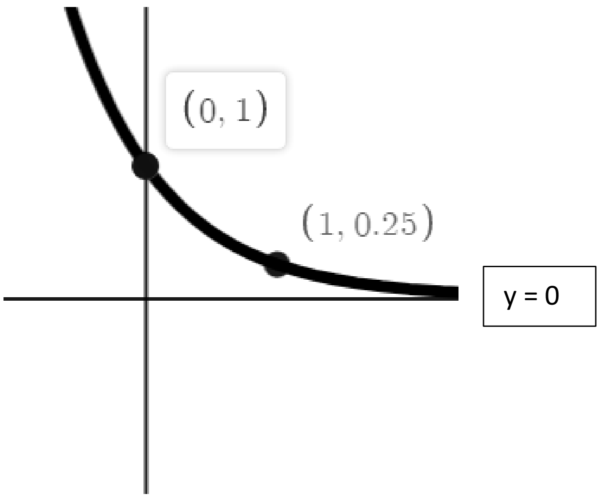


10. $y = -\log_3 x$

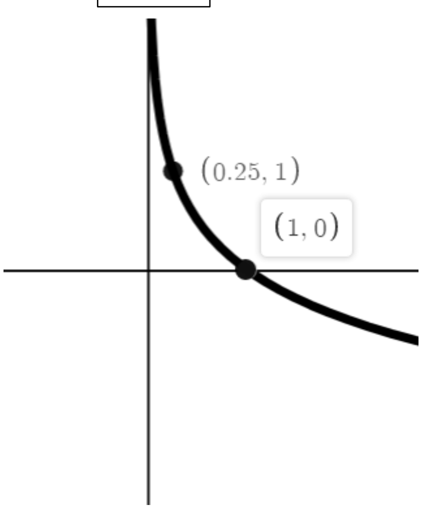
11.



12. $g(x) = -\log_2 x$
13. $g(x) = \log_2(-x)$
14. $y = \log_3 x$
15. $Y = \log_2(x+1) + 2$
16. $y = a^b$
17. x = 2
18. 2.58
19. 4.36
20. x = 2 or x = 0.63
21. $y = log_{\frac{1}{2}}x$
22. $y = 5^{-x}$ or you could say $y = \frac{1}{5}^x$
23. $y = -4^x$
24a.



24b. x = 0



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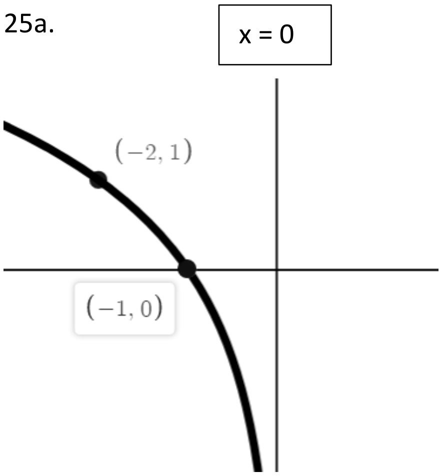
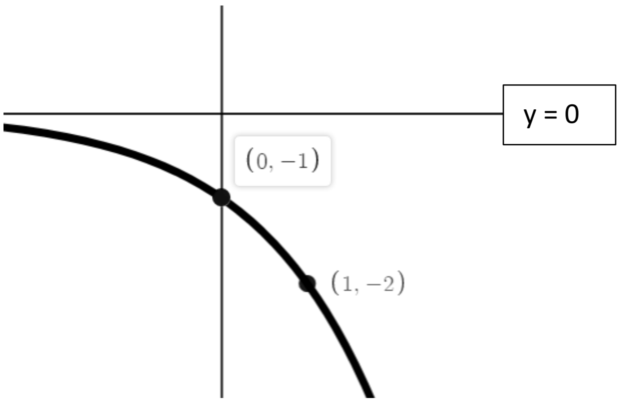


25b.



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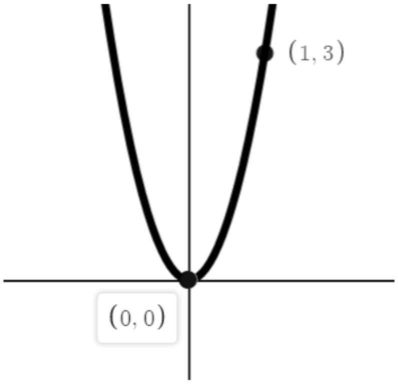

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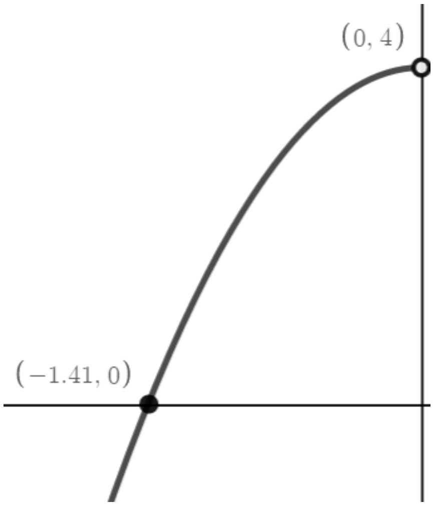
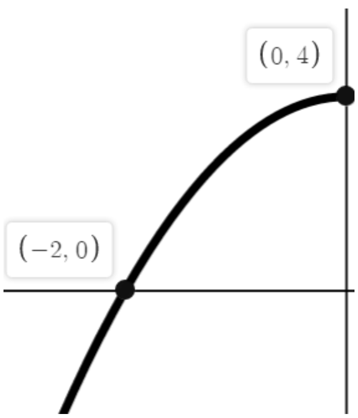


28. g(x) = -3x + 3
29. $g(x) = -\frac{2}{3}x^2$
30. $g(x) = -log_3(x-1)$
31.



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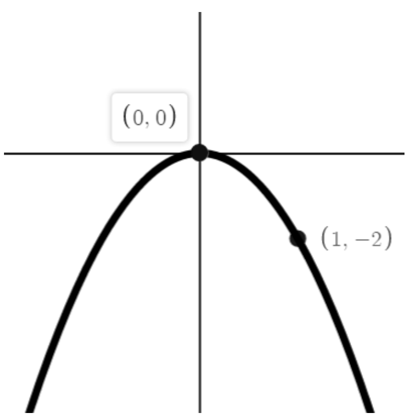

33. x = 9
34. x = 1.55
35. x = 1.74

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